Take-Home Task: Airbnb-Clone App

We'd like you to build a pixel-perfect clone of a real Airbnb listing page. The goal is to assess how precisely you can reproduce a polished, production-quality UI — exactly as it looks and exactly as it behaves — using AI-assisted development.

Reference (single source of truth)

https://airbnb-orpin-pi.vercel.app/

Your clone must match this reference exactly across:

- **Visual design:** identical layout, spacing, typography, colours, icons, and assets.
- **Animations and motion:** matching hover and scroll animations, and transitions.
- **Interaction and accessibility:** keyboard navigation, focus management, and accessibility.

AI-Native Development

We are looking for AI-native developers. Internally, we use tools like Claude Code, Codex, and Cursor heavily in our workflow, and we expect you to use AI-assisted development for this task too.

Please use the latest premium AI models available to you for free or at a discount (e.g. via the GitHub Student Developer Pack). With the right approach, **the UI portion of this task should take around 6 hours.**

Any direct lift-and-shift of the reference site's code (e.g. copying its bundled source/assets instead of reproducing it) may result in a lower score or disqualification.

What to Build

Recreate the Airbnb listing page and its two overlay views exactly as they look and behave on the reference. There are three views:

Screen	Expected Implementation
Listing Page	The full property page. Match the layout, spacing, typography, colours, and interactions exactly.
Photo Tour	Full-screen photo gallery opened from “Show all photos” or any hero image.
Lightbox	Single-photo viewer opened from any gallery photo, with prev/next arrows and keyboard ←/→ navigation. Match interactions and animations exactly.

Note: *Implement Desktop only — mobile is not required.*

Tech Stack

- **Frontend (required):** React (your MERN stack) — this is the primary focus and where most of the evaluation weight sits.
- **Backend + Database (bonus, optional):** A working Node/Express backend connected to MongoDB or PostgreSQL is a **bonus** — not required for a complete submission, but will earn extra credit if included (e.g. serving listing data via an API instead of hardcoding it in the frontend).
- No DB or booking/reservation logic is required — this task is visual/frontend-focused. If you do add a backend, read-only listing data served via API is sufficient; no write/booking flow needed.
- Deploy on Vercel or any free hosting platform of your choice.

Architecture Diagram

Submit a high-level architecture diagram for a **production-scale vacation-rental marketplace** (think Airbnb at scale). The diagram should illustrate your scaling strategy for:

- Frontend
- Backend
- Storage/Database
- Search
- Deployment

Tools like lucid.app or excalidraw are recommended. Submit as an image or PDF.

What We Will Look At

1. Modern AI workflow usage (coding agents, sub-agents, skills, prompts)
2. How closely the clone matches the reference — visual fidelity and behavioural parity, including animations, transitions, and accessibility
3. Production architecture thinking
4. AI sub-agent configs used for code quality and project structure
5. **Bonus:** a working backend + database integration

Please keep scope focused — a clean, complete frontend implementation is better than an over-engineered, incomplete one. Include your sub-agent/skill config files in your submission.

Submission

Important: Do not push the code to a public GitHub repository.

Submit as instructed over email. Your submission must include:

1. A zipped file containing the code and the architecture diagram (image/PDF file).
2. The sequence of prompts used for AI-assisted development (may be requested during submission/interview).